

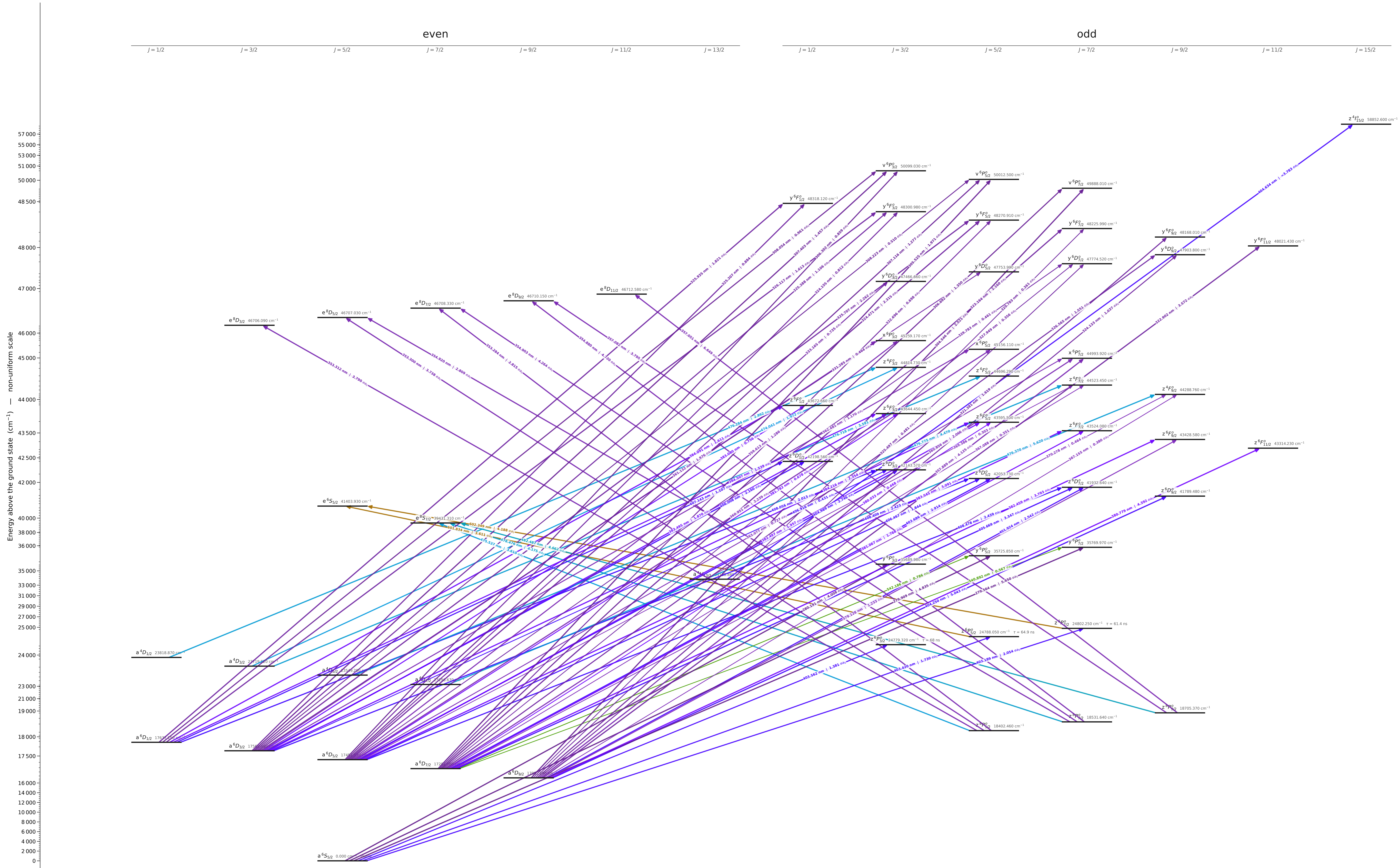
Manganese

Mn (Mn I). Strongest classified transitions below 2 μm (NIST): 59 levels, 90 lines.

Mn ⁺ ionisation limit 59959.56 cm⁻¹ = 7.43404 eV (166.779 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U|er|U' \rangle$ in ea.u.
Matrix elements are measured or NIST values; * = high-accuracy theory; ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er U' \rangle$ (e.a.u.)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁶ S _{5/2} - y ⁶ P _{3/2}	279.5641	279.4817	1072.3567		5.650	3.700e+08	-	- 279.5 nm strong UV resonance line a6S5/2 - y6P7/2
a ⁶ S _{5/2} - y ⁶ P _{5/2}	279.9094	279.8269	1071.0340		4.835	3.600e+08	-	- 279.8 nm strong UV resonance line a6S5/2 - y6P5/2
a ⁶ S _{5/2} - y ⁶ P _{7/2}	280.1907	280.1081	1069.9587		4.008	3.700e+08	-	- 280.1 nm strong UV resonance line a6S5/2 - y6P3/2
a ⁶ S _{5/2} - z ⁶ P _{1/2}	403.1892	403.0753	743.5527		2.054	1.630e+07	2.594	100 403.1 nm resonance line a6S5/2 - z6P7/2
a ⁶ S _{5/2} - z ⁶ P _{3/2}	403.4202	403.3062	743.1270		1.730	1.540e+07	2.451	100 403.3 nm resonance line a6S5/2 - z6P5/2
a ⁶ S _{5/2} - z ⁶ P _{5/2}	403.5623	403.4483	742.8653		1.381	1.470e+07	2.339	100 403.4 nm resonance line a6S5/2 - z6P3/2
a ⁴ F _{3/2} - z ⁴ F _{5/2}	404.6335	404.5193	740.8987		≈3.783	2.735e+07	-	- laser-cooling

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁵⁵Mn hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
a ⁶ S _{5/2}	-71.950		F=0: +629.56 F=1: +557.61 F=2: +413.71 F=3: +197.86 F=4: -89.94 F=5: -449.69
a ⁴ D _{5/2}	509.650	131.910	F=2: -6947.23 F=3: -5457.85 F=4: -3456.63 F=5: -930.36 F=6: +2137.43 F=7: +5766.54
a ⁴ D _{3/2}	458.680	20.985	F=1: -5148.91 F=2: -4237.54 F=3: -2868.25 F=4: -1038.33 F=5: +1255.82 F=6: +4018.70
a ⁴ D _{5/2}	437.700	-44.969	F=0: -3861.35 F=1: -3412.86 F=2: -2519.92 F=3: -1190.63 F=4: +562.86 F=5: +2724.38
a ⁴ D _{3/2}	470.670	-65.954	F=1: -2517.18 F=2: -1523.08 F=3: -81.39 F=4: +1748.52
a ⁴ D _{1/2}	881.390		F=2: -1542.43 F=3: +1101.74
z ⁶ P _{3/2}	572.600	11.992	F=1: -2997.76 F=2: -1862.15 F=3: -149.75 F=4: +2150.25
z ⁶ P _{5/2}	467.680	-74.948	F=0: -4144.66 F=1: -3659.00 F=2: -2694.41 F=3: -1264.38 F=4: +610.83 F=5: +2904.26
z ⁶ P _{7/2}	428.700	62.956	F=1: -4789.15 F=2: -3949.74 F=3: -2683.87 F=4: -983.46 F=5: +1162.29 F=6: +3766.86
y ⁶ P _{3/2}	-971.330	17.988	F=1: +5112.07 F=2: +3155.02 F=3: +232.94 F=4: -3637.99
y ⁶ P _{5/2}	-539.630	-68.952	F=0: +4673.50 F=1: +4150.41 F=2: +3098.05 F=3: +1503.98 F=4: -650.40 F=5: -3389.93
y ⁶ P _{7/2}	-389.730	-185.870	F=1: +4284.89 F=2: +3558.53 F=3: +2449.09 F=4: +932.65 F=5: -1022.63 F=6: -3456.61
z ⁴ D _{5/2}	86.940		F=2: -1195.42 F=3: -934.61 F=4: -586.85 F=5: -152.15 F=6: +369.50 F=7: +978.08
z ⁴ D _{3/2}	38.973		F=1: -438.45 F=2: -360.50 F=3: -243.58 F=4: -87.69 F=5: +107.18 F=6: +341.01
z ⁴ D _{1/2}	-26.981		F=0: +236.08 F=1: +209.10 F=2: +155.14 F=3: +74.20 F=4: -33.73 F=5: -168.63
z ⁴ F _{3/2}	-149.900		F=1: +786.98 F=2: +487.18 F=3: +37.48 F=4: -562.12
z ⁴ F _{5/2}	-821.430		F=2: +1437.50 F=3: -1026.79
z ⁴ F _{7/2}	179.880		F=3: -2923.05 F=4: -2203.53 F=5: -1304.13 F=6: -224.85 F=7: +1034.31 F=8: +2473.35
z ⁴ F _{9/2}	155.890		F=2: -2143.49 F=3: -1675.82 F=4: -1052.26 F=5: -272.81 F=6: +662.53 F=7: +1753.76
z ⁶ F _{7/2}	137.900		F=1: -1551.38 F=2: -1275.58 F=3: -861.88 F=4: -310.27 F=5: +379.23 F=6: +1206.62
z ⁶ F _{9/2}	173.880		F=0: -1521.45 F=1: -1347.57 F=2: -999.81 F=3: -478.17 F=4: +217.35 F=5: +1086.75
z ⁶ F _{11/2}	272.810		F=1: -1432.25 F=2: -886.63 F=3: -68.20 F=4: +1023.04
z ⁶ F _{13/2}	131.910		F=2: -1813.76 F=3: -1418.03 F=4: -890.39 F=5: -230.84 F=6: +560.62 F=7: +1483.99
z ⁶ F _{15/2}	170.880		F=1: -1922.40 F=2: -1580.64 F=3: -1068.00 F=4: -384.48 F=5: +469.92 F=6: +1495.20
x ⁶ P _{1/2}	281.800		F=1: -3170.25 F=2: -2606.65 F=3: -1761.25 F=4: -634.05 F=5: +774.95 F=6: +2465.75
x ⁶ P _{3/2}	290.800		F=0: -2544.50 F=1: -2253.70 F=2: -1672.10 F=3: -799.70 F=4: +363.50 F=5: +1817.50
x ⁶ P _{5/2}	374.740		F=1: -1967.38 F=2: -1217.90 F=3: -93.69 F=4: +1405.28

Measured A, B constants: sources in the data file.